ROAD ACCIDENT DATA ANALYSIS

Name:

Dikshya

Panda

Student ID:

AF05219426

Batch Code:

ANP-D4853

Project Name:

Road Accident Data Analysis using Power BI

Project Guide:

Ms. Rajshri Thete

Objective

The objective of Road Accident Analysis is to extract meaningful insights, identify accident patterns, and visualize road safety statistics using Microsoft Power BI. The project focuses on understanding accident severity, vehicle involvement, road conditions, light conditions, and yearly trends to support better decision-making for road safety.

Exploratory Analysis

- Understanding road accident trends across different years.
- Comparing fatal, serious, and slight casualties.
- Analyzing accident distribution by vehicle type.
- Studying the impact of light conditions and road surfaces.
- Examining accident occurrence in urban and rural areas.

Data Visualization

- Representing accident statistics using KPI Cards, Pie Charts, Donut Charts, Bar Charts, and Line Charts.
- Creating an interactive dashboard for easy analysis.
- Comparing accident patterns through visual reports.

Performance Analysis

- Identifying the vehicle type with the highest casualties.
- Comparing casualties under different road surface conditions.
- Evaluating urban and rural accident statistics.
- Studying monthly casualty trends for 2021 and 2022.

Dashboard Development

- Building an interactive dashboard using Microsoft Power BI.
- Using Power Query for data cleaning and DAX for calculated measures.
- Creating user-friendly visual reports for better interpretation.

Problem Statement

Road accidents generate a large amount of data related to casualties, vehicle types, road conditions, locations, and environmental factors. Analyzing this information manually is difficult and time-consuming.

This project aims to use Microsoft Power BI to analyze road accident data and present meaningful insights through an interactive dashboard by addressing the following challenges:

1. Importing and preprocessing accident datasets while ensuring data quality and consistency.

2. Analyzing casualty trends, accident severity, and environmental factors affecting accidents.

3. Visualizing accident statistics using KPI cards, charts, and dashboards to support better understanding of road safety.

Data Description

Source

The dataset was retrieved from **Kaggle** and contains historical Road Accident match information.

Data Model

A data model defines the structure and relationships within the Road Accident dataset, ensuring efficient storage, processing, and visualization of match statistics.

1. Data Model Overview

The data model follows a structured tabular format where each record represents an Road Accident data and contains information related to Casualties, vehicle type, Road Surface and light condition.

Entities and Attributes

Attribute Name		Description
Accident date		Date of accident occurrence
Casualties		Total number of casualties
Accident severity		Fatal, Serious, or Slight
Vehicle type		Type of vehicle involved
Road surface		Dry, Wet or snow/Ice
Light conditions		Daylight or darkness
Area		Urban or rural
Year		Year of accident

Road type		Type of a road where accident occurred

2. Data Relationships

- Every accident record contains one accident severity level.
- Each accident is associated with a vehicle type.
- Every accident occurs under a specific light condition.
- Each accident is linked to a road surface and road type.
- Every accident is classified as occurring in either an urban or rural area.

Approach

1. Data Import

Software Used

- Microsoft Power BI Desktop
- Power Query

Process

- Import the IPL CSV dataset.
- Verify column names and data types.
- Load the dataset into Power BI.

2. Data Cleaning

- Check for missing values.
- Remove duplicate records.
- Correct data types.
- Standardize categorical names.
- Prepare the dataset for visualization.

3. Data Analysis and Visualization

The dashboard includes:

- KPI Cards showing Total Casualties, Fatal Casualties, Serious Casualties, and Slight Casualties.
- Pie Chart displaying casualties by Light Condition.
- Bar Chart showing casualties by Vehicle Type.
- Donut Chart showing casualties by Area.
- Bar Chart displaying casualties by Road Surface.
- Column Chart showing Top 3 Casualties by Road Type.
- Line Chart comparing Monthly Casualties Trends (2021 vs 2022).

Project Result

1. Data Collection and Preparation

The Road accident dataset was imported from CSV files downloaded from Kaggle.

The dataset was cleaned using Power Query to remove inconsistencies and prepare it for analysis.

2. KPI Dashboard

Displays:

- Total Casualties: 418K
- Fatal Casualties: 3,953
- Serious Casualties: 41K
- Slight Casualties: 263K

Result: Provides a quick overview of accident severity statistics

3. Light Condition Analysis

Displays casualties under Daylight and Darkness.

Result: Most accidents occurred during daylight conditions.

4. **Vehicle Type Analysis**

Displays casualties by vehicle type.

Result: Cars contributed to the highest number of casualties.

5. **Area Analysis**

Displays accident distribution in Urban and Rural areas.

Result: Urban areas recorded a higher percentage of casualties than rural areas.

6. **Road Surface Analysis**

Displays casualties based on road surface conditions.

Result: Dry road surfaces accounted for the highest number of casualties.

7. **Road Type Analysis**

Displays the top three road types with the highest casualties.

Result: Single carriageways recorded the highest number of casualties.

8. **Monthly Trend Analysis**

Displays monthly casualty trends for 2021 and 2022.

Result: Helps identify seasonal patterns and compare yearly accident trends.

Conclusion

This Road Accident Analysis project provides an interactive overview of accident trends by analyzing casualty severity, vehicle types, road conditions, and environmental factors using Microsoft Power BI. The dashboard transforms raw accident data into meaningful visualizations that support easy interpretation and better decision-making.

The project demonstrates the practical application of Business Intelligence techniques such as data cleaning, transformation, DAX calculations, and dashboard development. Future enhancements may include predictive analysis, location-based accident mapping, and machine learning models for accident forecasting.